

Networking Task 02: Network Devices & IP Addressing

Part A: Network Devices Research

1. Router

Purpose

A router is used to connect different networks and allow devices to access the Internet.

How it Works

The router receives data from devices and checks where the data needs to go. It then sends the data through the most suitable path to reach its destination.

Real-World Usage

At home, a router connects smartphones, laptops, and other devices to the Internet through a single connection provided by the ISP.

2. Switch

Purpose

A switch helps multiple devices communicate within the same network.

How it Works

The switch identifies the intended device and sends data only to that device instead of sending it to everyone on the network.

Real-World Usage

In offices and computer labs, switches connect computers, printers, and servers to create an efficient local network.

3. Hub

Purpose

A hub is a simple device used to connect several computers on a network.

How it Works

Whenever the hub receives data from one device, it forwards that data to all connected devices, whether they need it or not.

Real-World Usage

Hubs were commonly used in older networks for connecting computers but are now mostly replaced by switches.

4. Access Point

Purpose

An Access Point allows wireless devices to connect to a network.

How it Works

It receives data from a wired network and broadcasts Wi-Fi signals so that wireless devices can connect and communicate.

Real-World Usage

Access Points are used in schools, colleges, hotels, and offices to provide Wi-Fi coverage over a larger area.

5. Firewall

Purpose

A firewall protects computers and networks from unauthorized access and cyber threats.

How it Works

It monitors network traffic and applies security rules to decide whether data should be allowed or blocked.

Real-World Usage

Organizations use firewalls to prevent hackers from accessing their systems and to secure sensitive information.

6. Modem

Purpose

A modem provides a connection between a user's network and the Internet service provider.

How it Works

It converts signals from the ISP into a form that computers and networking devices can understand and use.

Real-World Usage

A modem is commonly installed in homes and offices to provide Internet access through broadband, fiber, or cable connections.

Part B: IP Address Classification

- **192.168.1.10 – Private IP Address**
This IP belongs to the private range **192.168.0.0 to 192.168.255.255**, which is reserved for use within local networks.
- **10.0.0.5 – Private IP Address**
This IP falls within the private range **10.0.0.0 to 10.255.255.255**, commonly used in homes, offices, and organizations.
- **172.16.5.20 – Private IP Address**
This IP is part of the private range **172.16.0.0 to 172.31.255.255**, which is reserved for internal network communication.
- **8.8.8.8 – Public IP Address**
This IP does not belong to any private IP range and is accessible over the Internet, making it a public IP address.

- **1.1.1.1 – Public IP Address**

This IP is outside all private IP ranges and can be reached from anywhere on the Internet, so it is a public IP address.

- **192.168.100.1 – Private IP Address**

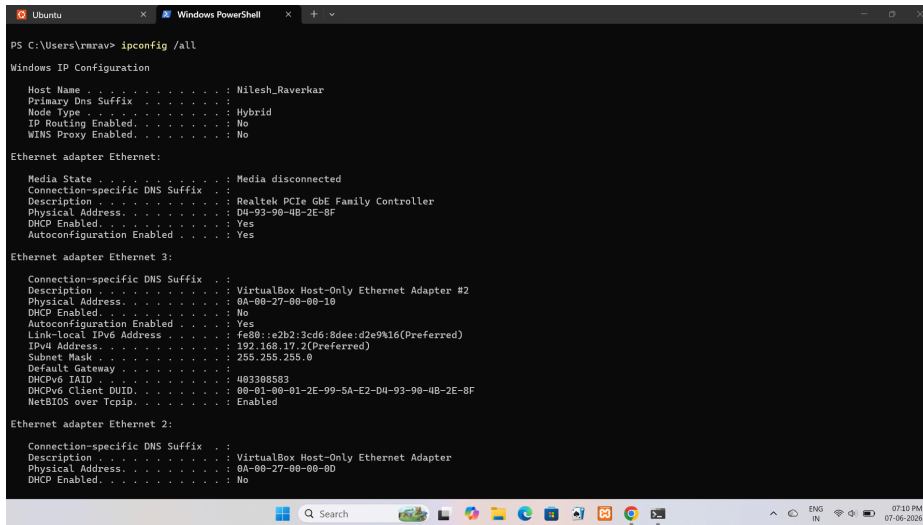
This IP falls within the private range **192.168.0.0 to 192.168.255.255**, making it a private IP address.

Part C: Understanding Your Network

How to find the details (Windows)

1. Press Windows + R
2. Type cmd and press Enter
3. Type:

`ipconfig /all`



```
PS C:\Users\rmrav> ipconfig /all

Windows IP Configuration

Host Name . . . . . : Nilesch_Raverkar
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 0A-00-27-00-00-10
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter #2
Physical Address. . . . . : 0A-00-27-00-00-10
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e2b2:3cd6:8dee:d2e9%16(Preferred)
IPv4 Address. . . . . : 192.168.17.2(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 403388583
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-99-5A-E2-04-93-90-00-00-00-00-00
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-00
DHCP Enabled. . . . . : No
```

1. Ipv4 address:

192.6.43.3

2. DNS Server:

172.31.123.241

3. Default Gateway:

fe80::8ca4:7dff:fe0f:91b4%15

1. Which IP range does your device belong to?

My device belongs to the 192.6.43.3 IP range.

2. Is it Public or Private?

It is a **Private IP address** because it is used within a local network and is not directly accessible from the Internet.

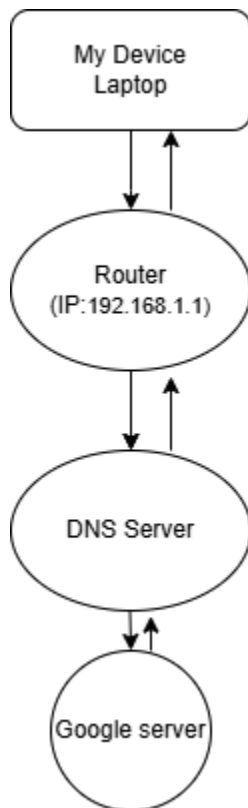
3. What role does your router play in your network?

The router acts as a gateway between my local network and the Internet. It forwards data packets, assigns IP addresses to devices, and enables communication with external networks.

4. What would happen if the DNS server stopped working?

If the DNS server stopped working, websites could not be accessed using their domain names (such as google.com). Internet access might still work using IP addresses, but users would have difficulty browsing because domain names would not be translated into IP addresses.

Part D: Network Communication Flow



Step-by-step explanation

1. My Device

I enter www.google.com in the browser. My device first checks how to find the correct IP address using DNS.

2. Router

The router forwards my request from the local network to the Internet. It works as a gateway between my device and the outside network.

3. DNS Server

The DNS server converts www.google.com into its IP address so the system can locate Google's server.

4. Google Server

The request reaches Google's server using the IP address, and it processes the request and prepares the webpage.

5. Response back to My Device

Google's server sends the response back through the network path, and the webpage loads on my device.

Part E: Practical Command Exercise

```
netbios over tcpip . . . . . Enabled
PS C:\Users\rmrav> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::e2b2:3cd6:8dee:d2e9%16
```

```
Ubuntu  Windows PowerShell
Ethernet adapter vEthernet (WSL (Hyper-V firewall)):
```

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::eb5f:bbf2:8e10:8af8%66
IPv4 Address. . . . . : 172.28.208.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :

PS C:\Users\rmrav> ping google.com

Pinging google.com [2404:6800:4009:806::200e] with 32 bytes of data:
Reply from 2404:6800:4009:806::200e: time=27ms
Reply from 2404:6800:4009:806::200e: time=46ms
Reply from 2404:6800:4009:806::200e: time=31ms
Reply from 2404:6800:4009:806::200e: time=32ms

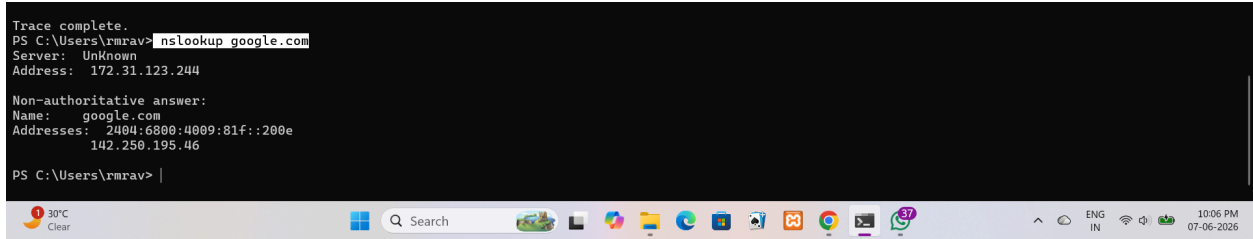
Ping statistics for 2404:6800:4009:806::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 27ms, Maximum = 46ms, Average = 34ms
PS C:\Users\rmrav> tracert google.com

Tracing route to google.com [2404:6800:4009:806::200e]
over a maximum of 30 hops:
  0  4 ms  2 ms  1 ms  2409:40c2:6407:dbd0:96
  1  *      *      *      Request timed out.
  2  35 ms  20 ms  19 ms  2405:208:5205:28:3925::ff22
  3  *      *      *      Request timed out.
  4  *      *      *      Request timed out.
  5  *      *      *      Request timed out.
  6  39 ms  28 ms  23 ms  2405:208:801:6000:10
  7  |
```

```
Trace complete.
PS C:\Users\rmrav> nslookup google.com
Server: UnKnown
Address: 172.31.123.244

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:81f::200e
          142.250.195.46

PS C:\Users\rmrav> |
```



1.What IP address did DNS return for Google?

DNS returned a public IP address of Google (example: 142.250.192.46).

2.Was the ping successful?

Yes, the ping was successful.

3.Why is DNS important before communication begins?

DNS is important because it converts website names into IP addresses so devices can communicate.